

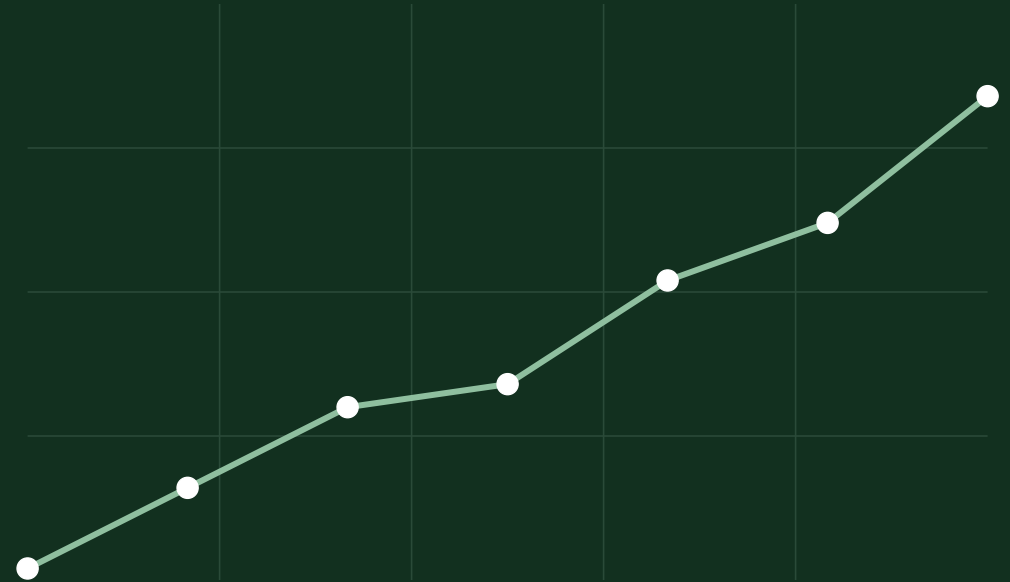
SIGNAL

Developmental screening for children in institutional care

Every observation documented. Every flag explained.

Sami · Ayesha · Amaan

AI Hackathon Pakistan · A screening aid, not a diagnostic tool



Two structural failures, not one

Speech, language and hearing delays are most treatable when caught early. In institutional care, two things make that almost impossible.

01

Nobody owns the observation

A caretaker notices something, mentions it, and it is gone. No record means nothing happens — and nothing can be shown to have happened.

02

Nobody knows how old they are

Every milestone chart assumes you know the child's age. For most children in institutional care, nobody does.

Who this affects

Children aged 0–6 in orphanages, shelters and welfare homes across Pakistan — and the untrained caretakers responsible for them.

*“An undocumented concern is not protection.
It is documented negligence.”*

Screening a child whose age nobody knows

Every developmental tool on the market takes a date of birth as a given. Ours cannot. So age is modelled as uncertainty rather than guessed at.

Confirmed date of birth

Milestones evaluate normally. Grades stand as computed.

Estimated age range

Evaluated at the younger bound. Age-dependent grades are downgraded automatically.

The rule that makes it safe

Uncertainty may only ever lower a grade — never raise one. But red flags that do not depend on age at all (a child losing words they already had; a caretaker's own hearing concern) stay exactly where they are, and every result carries an age-uncertain marker the clinician can see.

From a spoken sentence to a cited record

“He doesn't turn around when I clap behind him.”

01

Observation agent

Caretaker speaks or types, in Urdu or English. The agent extracts developmental signals — never a diagnosis.

02

Knowledge base

94 curated milestone and red-flag rows.
Speech/language and hearing retrieved JOINTLY — they present identically.

03

Reasoning agent

Asks adaptive follow-up questions, up to five turns, drawing each question from the matched clinical row.

04

Deterministic grade()

Rules decide the grade — not the model. HIGH · MODERATE · LOW-monitor · INSUFFICIENT.

05

Flag + reasoning trail

Every citation that produced the grade, snapshotted at the moment of writing.

Routed to a clinician, with a tamper-evident record of who noticed, when, and on what basis.

The model does not decide the outcome

This is the difference between a demo and something a clinician can act on. The LLM reads and asks. A pure rules function grades.

What the LLM does

- Extracts signals from free-text observation
- Asks the follow-up question the knowledge base supplies
- Writes the plain-language explanation

Never sees a name, an ID, or an institution.

What the code does

- Decides the grade from rules, deterministically
- Validates every citation against what was actually retrieved
- Refuses free recall — an invented citation raises an error

Same input, same grade. Every time.

Built to survive review, not just a demo

Hearing and language, never separately

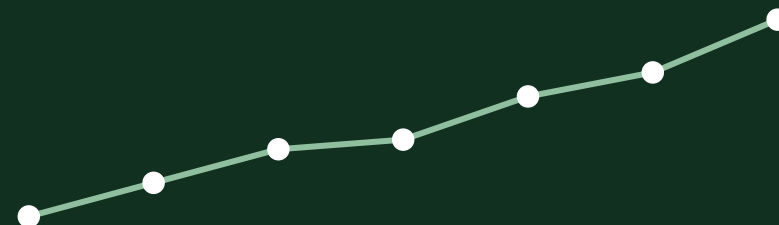
Hearing loss and language delay present almost identically. Screening one domain at a time is how children get misattributed. Both are retrieved together, and the domain is decided only after the joint check.

The trail is a snapshot, not a lookup

A clinician reading a flag six months later sees the exact wording the decision was made on — and is told if the reference has changed since. Editing a milestone cannot rewrite history.

Caretaker text is data, never instructions

Every observation is fenced as data in the prompt, and the system prompt forbids treating it as an instruction. A caretaker — or anyone typing into that box — cannot steer the screening. The outcome still comes from grounded clinical evidence.



Built for a regulated domain

Backend	FastAPI · SQLAlchemy 2 · Alembic · Python 3.12
Database	PostgreSQL 16 with row-level security for tenant isolation
Frontend	Next.js 16 (App Router) · React 19 · TypeScript · Tailwind 4
AI pipeline	Three narrow agents, strict JSON contracts — no framework, no embeddings
LLM	Any OpenAI-compatible endpoint · deterministic keyless reasoner for demos
Speech-to-text	Provider seam (Azure / Knowlez) — voice degrades to text, never blocks
Auth	RS256 JWT in an HttpOnly cookie — never localStorage
Secrets	Fernet-encrypted at rest · write-only admin API

CI runs pytest, bandit, pip-audit, npm audit and gitleaks on every pull request.

The database enforces it, not the code



Tenant isolation

Postgres row-level security — enforced by the database, not by remembering a WHERE clause.



Least privilege

The app connects as a role with no superuser, no BYPASSRLS and no DELETE grant.



Audit trail

Hash-chained and append-only. It records reads as well as writes, so “who opened this child's record” is answerable.



IDOR

A uniform 403 for both “does not exist” and “belongs to another institution” — existence never leaks.



No PHI in prompts or logs

The model receives age context and the observation. Never a name, an ID or an institution.



Nothing is ever deleted

Children archive with a mandatory reason; staff deactivate. Both reversible, both audited.

Row-level security is verified from the attacker's side: a test queries with no application filter at all, and the database still returns nothing.

FEASIBILITY

What we have actually built

This is a running system, not a prototype. It stands up from an empty checkout with three commands.

523

tests passing, against real
PostgreSQL — SQLite is never a
fallback

94

curated clinical rows from ASHA,
CDC, NIDCD and JCIH

13

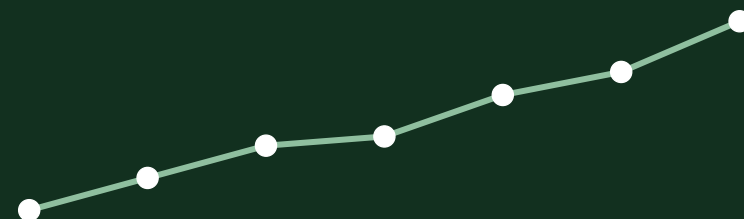
features delivered — the entire
Phase-1 backlog

0

High or Critical findings from bandit,
pip-audit, npm audit and gitleaks

Verified end to end

Login → intake with an estimated age → a voice or typed session → the adaptive reasoning
loop → a graded flag with its cited basis → referral → audit chain intact.



One test is failing on purpose

The R14 divergence

One acceptance case expects MODERATE; our counting rule computes HIGH. The three cited findings — “says what? often”, “sits close to the TV”, “recurrent ear infections” — are not three independent signals. They are one clinical picture: fluid behind the eardrum.

Resolving it is a clinical decision, not an engineering one. So it is documented and blocked rather than quietly patched to make the number look better.

Known limitations — none of them hidden in the code

- Password verification is not implemented. The build is hard-gated to synthetic data and refuses to run otherwise.
- No consent model and no retention policy — both need a legal decision before they need code.
- Access is institution-scoped, not relationship-based. Clinical practice would narrow it further.
- Rate limiting is in-process: correct for one instance, and Redis is the documented path beyond that.

The blockers are decisions, not code



Now

Clinical sign-off on R14, then flag clustering so correlated findings stop counting as independent ones.



Next

Password verification, a consent model, and a retention and erasure policy.



Then

Relationship-based access — scope a caretaker to the children actually assigned to them.



Scale

Redis-backed rate limiting and a managed key store, so the system runs on more than one instance.

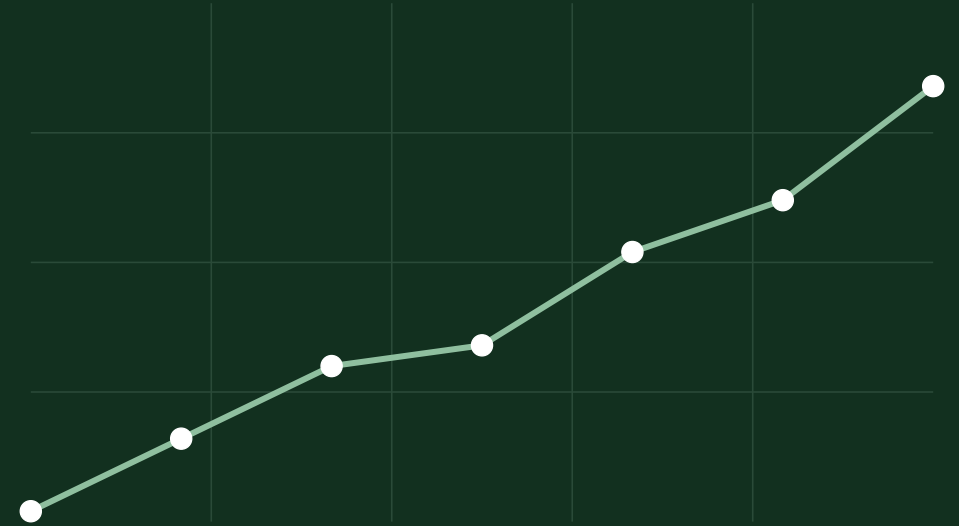
Nothing on this list is blocked on the model. Every one of them is blocked on a decision somebody has to make about children's data — which is the correct place for it to be blocked.

SIGNAL

The observation becomes a record.

The record has a response.

The response can be proven.



Team

Sami

Team lead

Ayesha

Clinical knowledge base

Amaan

Engineering

github.com/Amaan-Masood-0086/SIGNAL_dot_agent